Welcome to Section!!!

Week One

Slides created by Cameron Mohne and Maggie Lee

Introductions

A Little About Me

- My name is **ADEEBA RAFI**. I’ll be your Section Leader for Code in Place, and everyone here will be your section-mates!
- I a graduate student in Computer Science.
- In my free time I like to help others and teach.
- I enjoy playing basketball & badmention.

What About You All?

Go ahead and share:

1. Your name
2. Where you're tuning in from
3. One thing you're looking forward to (it doesn't have to be from Code in Place)!

If you aren't warmed up and comfortable with speaking just yet, that's fine! You can share directly to everyone in the chat or you can message me and I can read out your introduction for you!

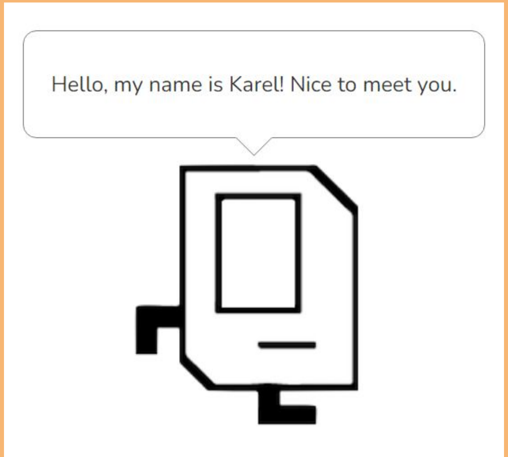
Another question: what can I do to make you feel included? Please feel free to private message me in the chat if there's anything I can do to make you more comfortable.

Breaking the Ice

- Share your names !!!
- Icebreaker Question: What's your dream project?
- If no one wants to share first, the person who is geographically closest to Stanford shares first!

Let's review and refresh these concepts a bit!

Karel Overview



- Karel is a small, but mighty robot!!!
- It has a few basic commands including: move(), turn_left(), pick_beeper(), and put_beeper()
- On its own, Karel has limited functionality, but with the help of our code, we can make great things happen!

Functions Overview

A function is a reusable block of code that does a specific task.

Each function has a specific purpose which breaks down a larger problem into smaller chunks—just like steps in making pasta from scratch!

To make a function, you need to define it using the **def** keyword. Afterwords, write the code you want the function to run. Make sure the code is indented below the function name like so:

```
def function_name():
    # Function code goes here!!!
```

Control Flow Overview

Lastly, let's take a look at control flow! We have:

1. Conditions (if statements)

- Tests for truth. Performs a block of code only when evaluated to true.

2. Loops (repeating)

- Performs some block of code, a specific amount of times.

3. While loop:

- Continuously perform a block of code until what's being tested is evaluated to false.

If-Statement

Used to make decisions.

An example if-statement that you may see and use with Karel:

```
def safe_move():
    if front_is_clear():
        move()
```

An if-statement runs code inside of it when the associated statement is evaluated to true. We will get into more complex statements later on in the course to add much more flexibility to our if-statements!

For-Loop

We use a for loop when we know how many times we want to repeat something.

An example for-loop that you may see and use with Karel:

```
def turn_right():
    for i in range(3):
        turn_left()
```

This loop is also called a *definite loop* because we know where it ends, when i reaches 3. (Be careful to remember that i begins at 0 when we start our loop!!!)

While-Loop

We use a while loop when we do not know how many times, but we have a condition.

An example while-loop that you may see and use with Karel:

```
def move_to_wall():
    while front_is_clear():
        move()
```

This loop is also called an *indefinite loop* because it will run until the associated condition becomes false, which may be never! Who knows? You will, hopefully. Be careful so you don't get stuck in an infinite loop while using this!

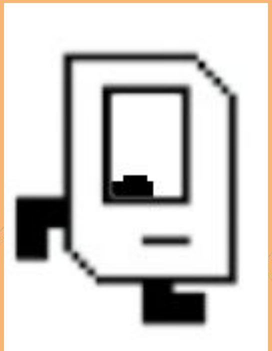
Lecture Review

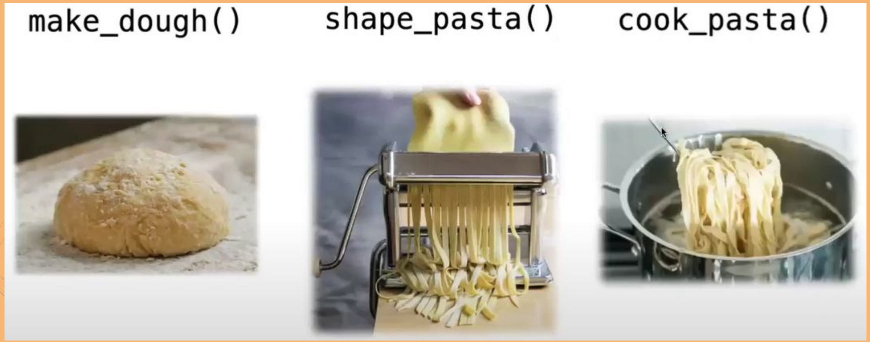
What We've Learned

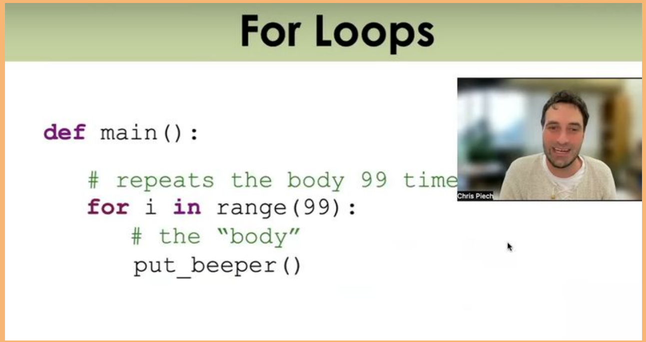
Before we get into our sample problem for today, let's review a bit. We've learned:

- The basics about Karel, the magnificent and wonderful robot
- Functions, a way of breaking down big problems into smaller chunks
- Control Flow, loops and conditional statements which guide our programs

This is a LOT of content, especially if you are newer to CS!







Any Questions?

Why are we here?

Section is what makes Code in Place different.

`section != lecture`

Building a community of learners.

How can you get the most out of this time?

This is your time.

Finishing problem isn't important.

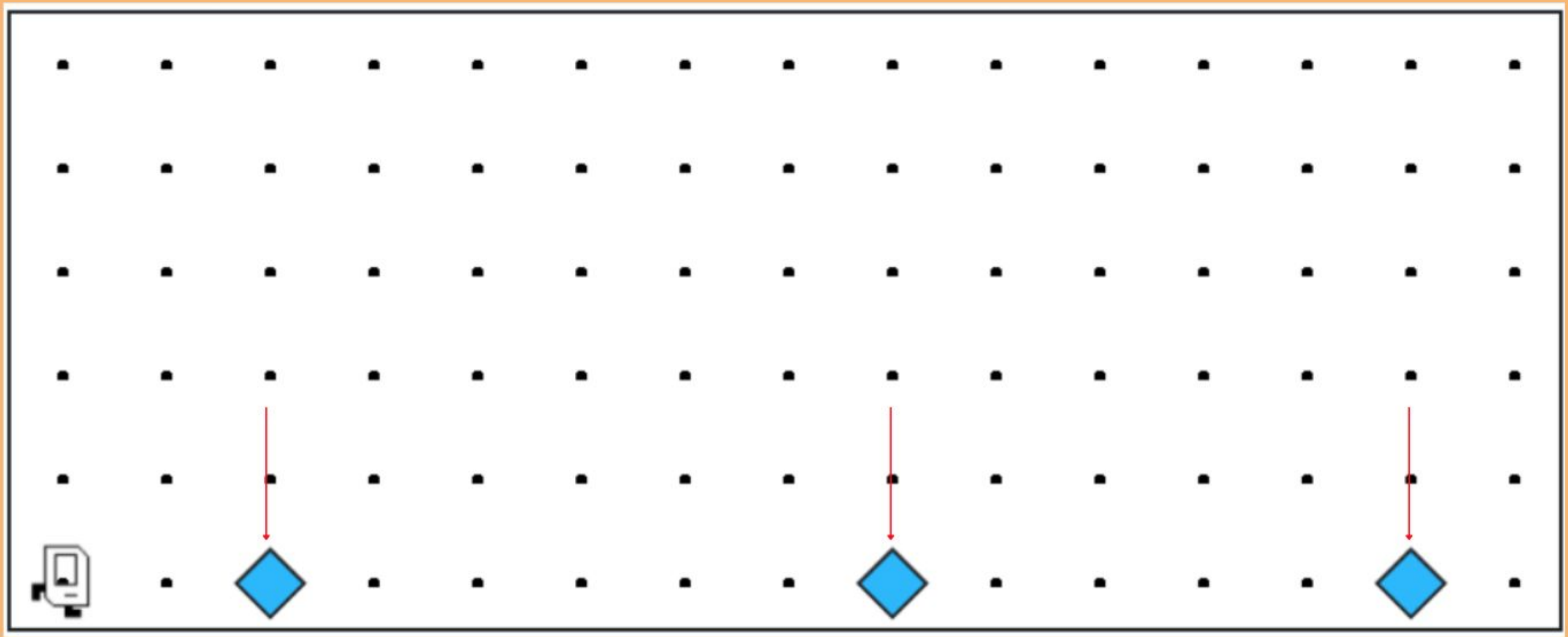
If we run over, no pressure to stay.

Section Problem: Hospital Karel

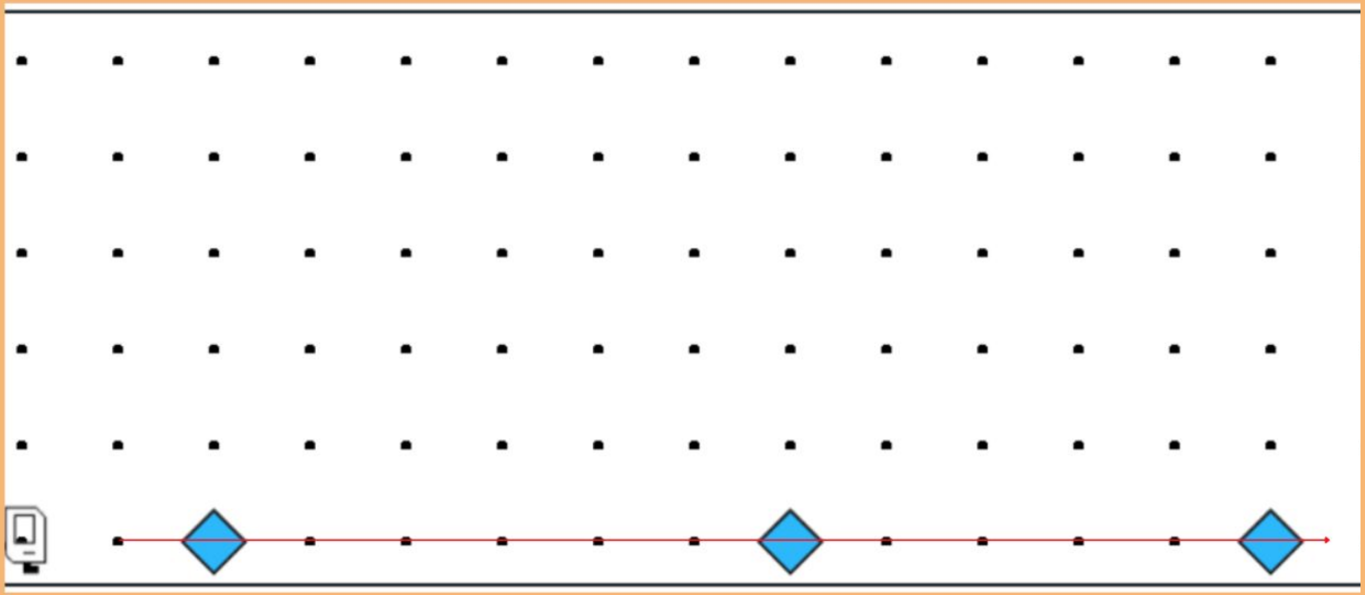
Setting Context

Countries around the world are dispatching hospital-building robots to make sure anyone who gets sick can be treated. They have decided to enlist Karel robots. Your job is to program those robots.

Each beeper in the figure represents a pile of supplies.



Karel's job is to walk along the row and build a new hospital in the places marked by each beeper.



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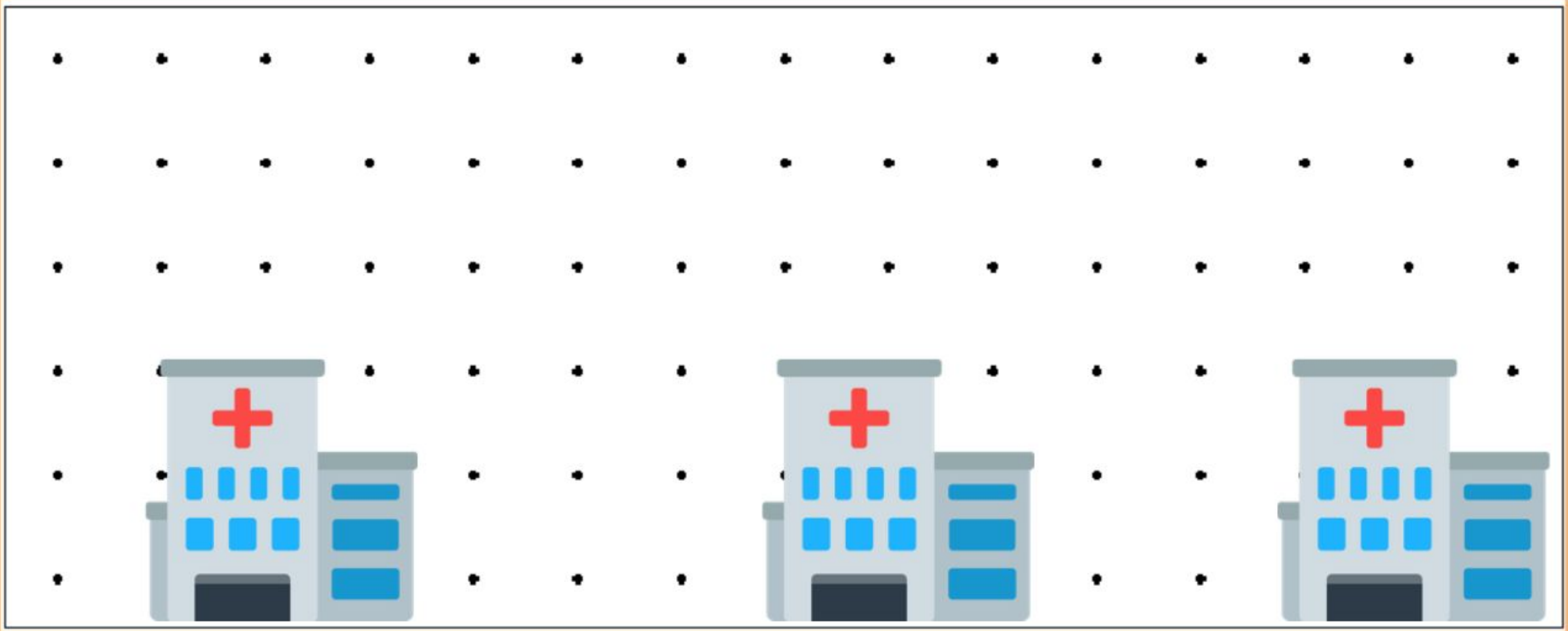
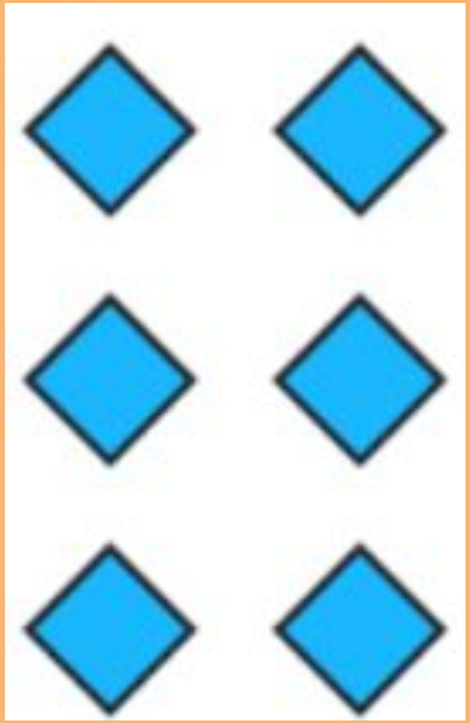
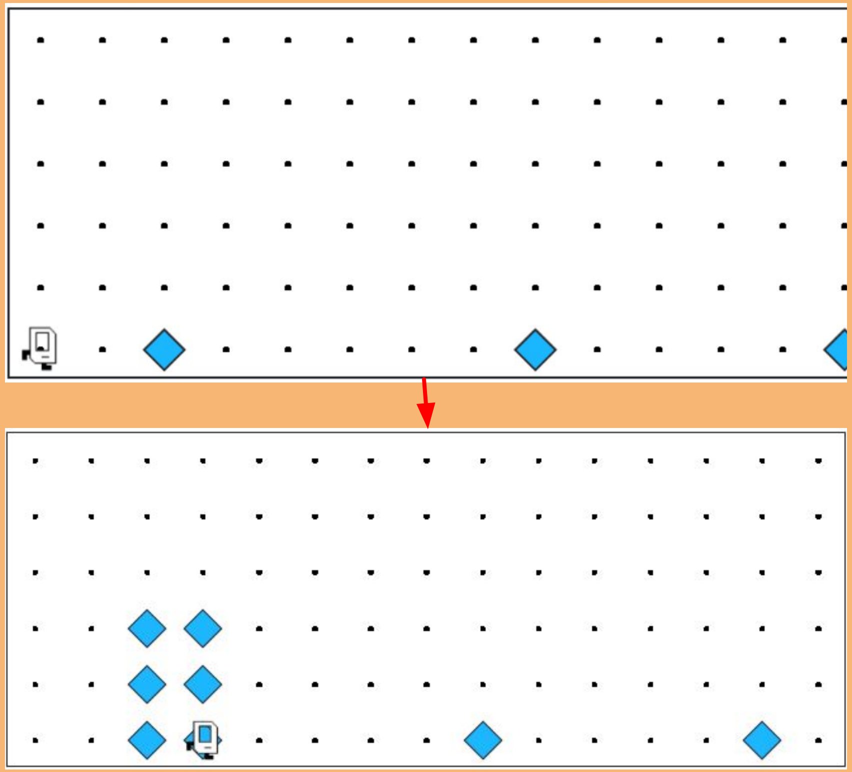


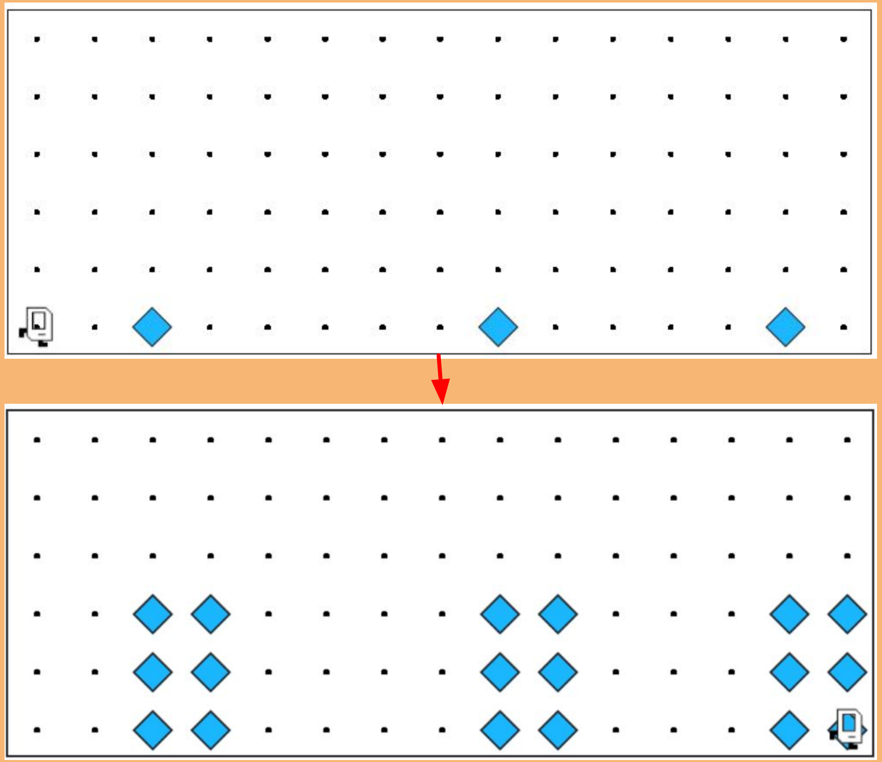
Hospitals look like this: a 3x2 rectangle of beepers!



The new hospital should have their corner at the point at which the pile of supplies was left.



At the end of the run, Karel should be at the end of the row having created a set of hospitals. For the initial conditions shown, the result would look like this:



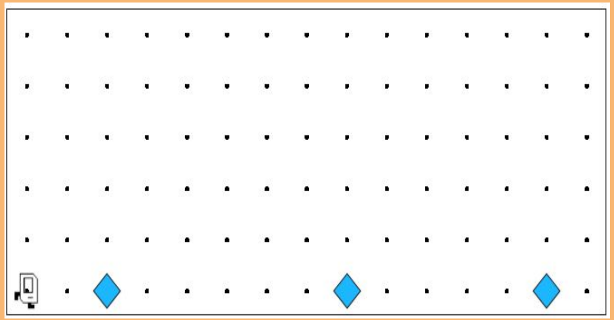
Notes to Keep in Mind

- Karel starts facing east at (1, 1) with an infinite number of beepers in its beeper bag.
- The beepers indicating the positions at which hospitals should be built will be spaced so that there is room to build the hospitals without overlapping or hitting walls.
- There will be no supplies left on the last column.
- Karel should not run into a wall if it builds a hospital that extends into that final corner.

Questions Before We Begin?

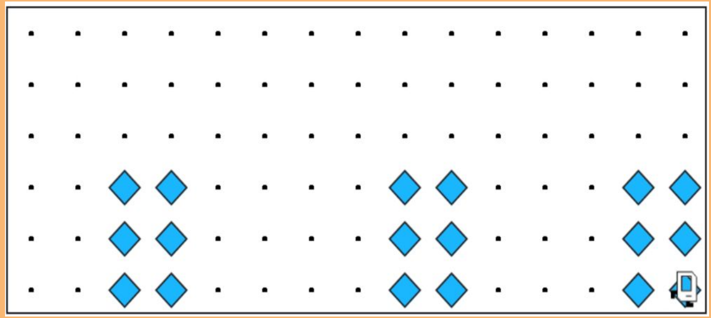
Start

While front is clear:
If there is a beeper:
Pick the beeper



Build first column:
Turn left
Put 3 beepers going up
Come back down
Turn to face right

Move right



Build second column:
Turn left
Put 3 beepers going up
Come back down
Turn to face right

Move forward

End